

ML Scientist with 1.5+ years building and shipping production AI end-to-end — real-time video analytics for McDonald's India, document processing at **100K+ documents/month**, and lead scoring on **8M+ behavioral sessions** — spanning Computer Vision, NLP, and distributed inference.

EDUCATION

HARCOURT BUTLER TECHNICAL UNIVERSITY

B.TECH IN COMPUTER SCIENCE & ENGINEERING
2021 – 2025 | Kanpur, India
CGPA: 8.3 / 10.0

DEWAN PUBLIC SCHOOL

SENIOR SECONDARY (XII)
2020 | Hapur, India
Score: 94%

DEWAN PUBLIC SCHOOL

SECONDARY (X)
2018 | Hapur, India
Score: 98%

LINKS

LinkedIn:// [paritkansal](#)
GitHub:// [ParitKansal](#)
Website:// [paritkansal.in](#)

SKILLS

LANGUAGES

Python • C++ • C

CORE EXPERTISE

Machine Learning • Deep Learning
Data Science • Computer Vision
NLP • Document AI
Real-Time Video Analytics
RAG • Graph Neural Networks
Few-Shot Learning • VLMs
Multimodal AI

ML & AI TECHNIQUES

CNNs • Transformers • LLMs
Statistical ML • Transfer Learning
Fine-Tuning • LoRA • QLoRA
Prompt Engineering • Feature Engineering
Ensemble Learning • Object Detection
Instance Segmentation

FRAMEWORKS

PyTorch • HuggingFace • TensorFlow
Scikit-learn • XGBoost • OpenCV
YOLO • LangChain • LangGraph
FastAPI • Triton Inference Server
NumPy • Pandas

INFRA & MONITORING

Docker • Git • CI/CD
MongoDB • MySQL • MinIO
Redis • Grafana • Loki • Power BI

CERTIFICATIONS

TENSORFLOW: ADVANCED TECHNIQUES

Coursera & DeepLearning.AI | Jun 2024

MACHINE LEARNING WITH PYTHON

freeCodeCamp | Apr 2024

EXPERIENCE

XELPMOC DESIGN AND TECH LTD | MACHINE LEARNING SCIENTIST

Jan 2025 – Present | Hyderabad, India

Docuxary – Intelligent Document Processing

- Architected an end-to-end IDP platform for invoices, receipts, and bank statements using **Gemini with extended thinking**, orchestrating multi-stage extraction, VLM-as-Judge verification, and QuickBooks sync via distributed async workers (RQ + Redis Streams)
- Built a **parallel multi-part extraction engine** that decomposes documents into schema-bound Pydantic sections, processes pages concurrently, and streams **real-time reasoning narratives** to clients via Redis pub/sub
- Designed a **VLM-as-Judge** verification layer performing sectional cross-checks between source document images and extracted fields, detecting **missing values, hallucinations, and mismatches** with per-section confidence scoring
- Integrated full **QuickBooks Online** automation with intelligent entity resolution, a preflight review gate, and idempotent create/update sync — **eliminating manual data entry** by automatically pushing extracted data into QuickBooks

Real-time Video Analytics (McDonald's India)

- Architected and deployed a scalable microservices-based real-time video analytics platform processing **multiple concurrent RTSP streams** for operational intelligence and compliance monitoring across stores
- Fine-tuned **YOLO, ViT + LoRA, InsightFace, and XCLIP** for staff availability monitoring, customer demographics, facial emotion recognition, table occupancy analysis, and behavioral compliance detection
- Implemented **zero-shot action recognition** pipelines using XCLIP to automatically detect policy violations including mobile phone usage, floor mopping, idle crowding, and staff engagement tracking
- Designed automated **hygiene and SLA monitoring** systems leveraging object detection, occupancy tracking, and anomaly detection for table cleanliness assessment, dirty-floor detection, and service-quality measurement
- Built a **fault-tolerant, low-latency** distributed architecture using Docker, Redis Streams, MinIO, MongoDB Replica Sets, Grafana, and Loki — enabling reliable multi-stream processing, full observability, and production-scale deployment

Intelligent Document Understanding (Firstsource)

- Fine-tuned **DONUT** for conditional key-value extraction reaching **98% field-level accuracy**
- Trained a **Gemma-3-4B** (4-bit VLM) automated judge to verify predictions, achieving **96.67% document-level accuracy** on **100K+ documents/month**

Multi-Channel Lead Scoring (ASBL)

- Built a multi-channel lead scoring pipeline fusing high-volume WhatsApp, web, and audio interaction data using **ModernBERT** and **XGBoost**, achieving **83% recall within the top 13% ranked events**
- Developed an end-to-end web session and visitor scoring system trained on **8M+ behavioral sessions**, achieving **88% session-level** and **90% visitor-level** recall

Document Detection Pipeline (ASK Business Services)

- Created a synthetic document-generation pipeline to train **Mask R-CNN** for multi-document detection, achieving **97.1% AP@[IoU=0.50:0.95]**

VODAFONE IDEA FOUNDATION | DATA ANALYTICS INTERN

Jun 2024 – Jul 2024 | Remote

- Applied EDA, statistical analysis, and **Lasso and Bayesian Regression** to quantify the impact of pricing, distribution, promotions, and competition on sales, generating actionable business insights
- Built an interactive **Power BI** dashboard for scenario-based sales forecasting and business decision support

ACADEMIC PROJECTS

COMPARATIVE ANALYSIS OF RECOMMENDATION ALGORITHMS | JUN 2024

Benchmarked SVD, SVD++, KNN, RBM, and AutoRec on RMSE, Hit Rate, and Novelty to produce side-by-side performance comparisons.

QUESTION PAIR SIMILARITY DETECTION | OCT 2023

Built a duplicate-question detection system on a **400K+ Quora dataset** (85.32% accuracy) by engineering 15+ token-overlap and fuzzy-matching features, and optimizing ensemble models (XGBoost, CatBoost, LightGBM, HistGradientBoosting) via hyperparameter tuning.